

The Linear Regression Model

Part II — Goodness of Fit, OLS Properties & Multiple Regression

Paulo Fagandini

Lisbon Accounting and Business School — Polytechnic University of Lisbon

Goodness of Fit

Lecture Overview

Lecture 9 — Goodness of Fit, OLS Properties & Multiple Regression

- How well does the fitted line describe the data? — the variance decomposition
- The coefficient of determination R^2
- The correlation coefficient r and its link to R^2
- Why OLS is a good estimator — the Gauss–Markov theorem (BLUE)
- What goes wrong when we omit a relevant variable — **Omitted Variable Bias**
- Extending to **Multiple Linear Regression (MLR)**
- Interpreting MLR coefficients
- Adjusted R^2

The Variance Decomposition

Decomposing Total Variation

For each observation i , we can write:

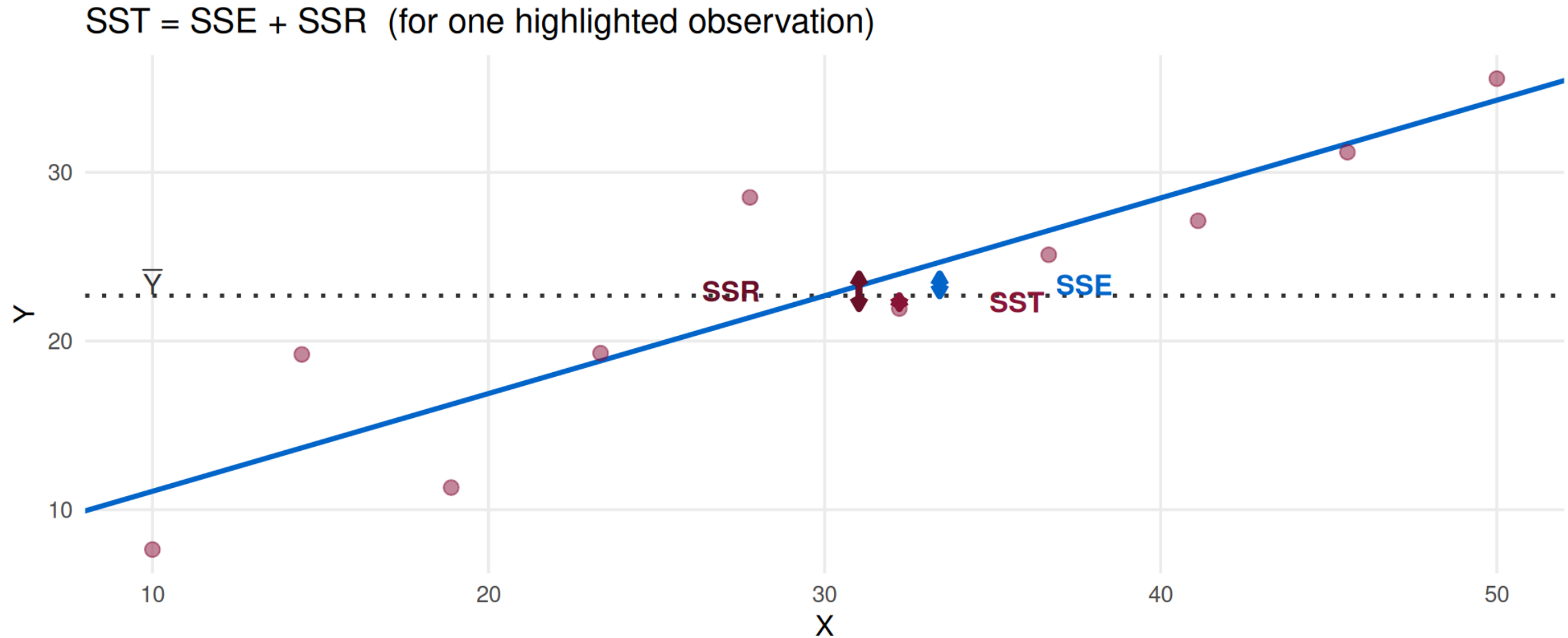
$$Y_i - \bar{Y} = \underbrace{(\hat{Y}_i - \bar{Y})}_{\text{explained by regression}} + \underbrace{(Y_i - \hat{Y}_i)}_{\text{residual}}$$

Squaring and summing over all observations:

$$\underbrace{\sum (Y_i - \bar{Y})^2}_{\text{SST}} = \underbrace{\sum (\hat{Y}_i - \bar{Y})^2}_{\text{SSE}} + \underbrace{\sum (Y_i - \hat{Y}_i)^2}_{\text{SSR}}$$

Term	Name	Meaning
SST	Total Sum of Squares	Total variation in Y
SSE	Statistics II — Linear Regression: Goodness of Fit & MLR Explained Sum of Squares	Variation explained by X

Visualising the Decomposition



The Coefficient of Determination R^2

Definition of R^2

The **coefficient of determination** measures the **proportion of total variation** in Y explained by the regression:

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

Key properties:

- $0 \leq R^2 \leq 1$
- $R^2 = 0$: X explains nothing
- $R^2 = 1$: perfect fit (all points on the line)

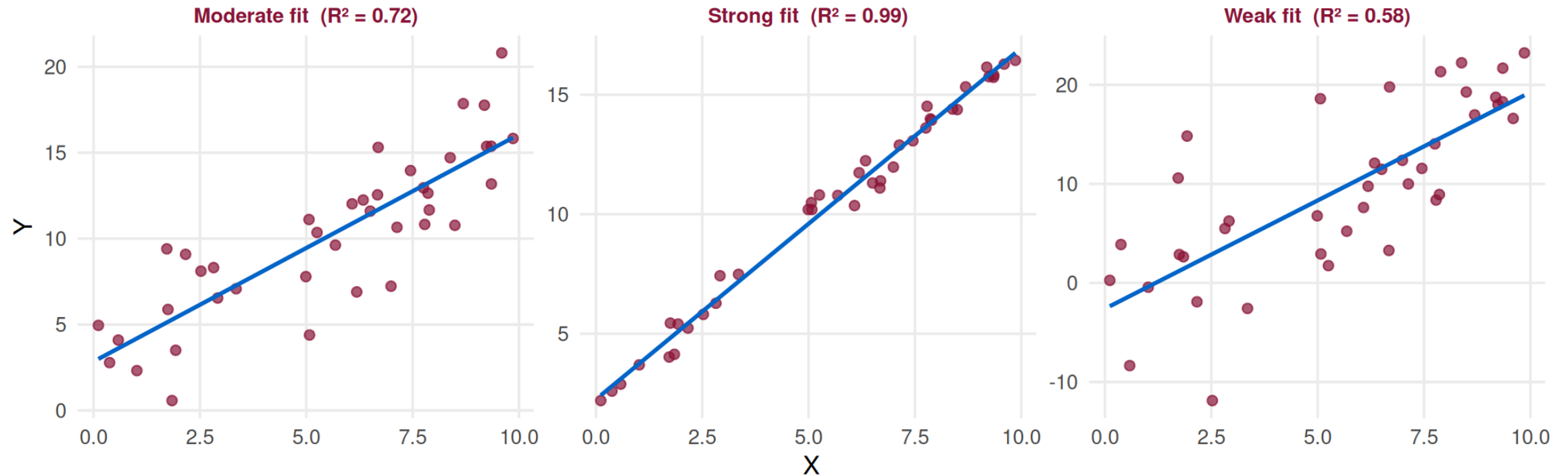
R^2 is unit-free

Interpretation:

“The model explains $R^2 \times 100\%$ of the total variation in Y .”

An R^2 of 0.82 means 82% of the variation in Y is explained by X via the estimated model

R^2 : Examples Across Different Fits



Caution: R^2 Has Limits

- A **high** R^2 does not mean the model is correctly specified — it could be driven by a single outlier, or by a spurious correlation.
- A **low** R^2 does not mean the regression is useless — in many economic applications, $R^2 = 0.3$ is perfectly reasonable.
- **Adding more regressors always increases R^2** — even if those variables are irrelevant. This is why MLR uses **Adjusted R^2** instead (covered shortly).
- **R^2 does not measure causality.** Two unrelated trending variables can produce $R^2 \approx 1$ (spurious regression).

⚠ Important

The Correlation Coefficient r

In SLR, the sample **Pearson correlation coefficient** is:

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \cdot \sum (Y_i - \bar{Y})^2}} = \frac{S_{XY}}{\sqrt{S_{XX} \cdot S_{YY}}}$$

Properties:

- $-1 \leq r \leq 1$
- Sign of r = sign of $\hat{\beta}_1$
- $r = 0$: no linear association
- $|r| = 1$: perfect linear fit

Link to R^2 :

In SLR (one regressor):

$$R^2 = r^2$$

The coefficient of determination equals the square of the correlation coefficient.

R^2 and r in R

Continuing with our running example:

```
1 # R-squared  
2 summary(model)$r.squared
```

```
[1] 0.5416654
```

```
1 # Correlation coefficient  
2 cor(revenue, profit)
```

```
[1] 0.7359792
```

```
1 # Verify:  $r^2 = R^2$   
2 cor(revenue, profit)^2
```

```
[1] 0.5416654
```

Interpretation: the model explains 54.2% of the total variation in operating profit. The correlation between revenue and profit is $r = 0.736$ — a strong positive linear association.

Why Trust OLS?

The Gauss–Markov Theorem

! Important

Gauss–Markov Theorem:

Under assumptions A1–A5, the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are **BLUE**:

- **Best**: minimum variance among all linear unbiased estimators
- **Linear**: linear functions of $Y_1, \dots, Y_n$
- **Unbiased**: $E[\hat{\beta}_j] = \beta_j$ for $j = 0, 1$
- **Estimator**

Unbiasedness means: if we repeated our sample many times, the average of the $\hat{\beta}_j$ values would equal the true β_j .

Minimum variance means: no other linear unbiased estimator can be more precise than OLS.

Unbiasedness of OLS

We can show that, under A1–A5:

$$E[\hat{\beta}_1] = \beta_1, \quad E[\hat{\beta}_0] = \beta_0$$

The intuition: OLS residuals are constructed to be uncorrelated with X (by the first-order conditions), so the error term ε_i does not “leak” into the estimates on average.

Variance of the slope estimator:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{XX}} = \frac{\sigma^2}{\sum (X_i - \bar{X})^2}$$

Omitted Variable Bias

What Happens When We Omit a Variable?

Suppose the **true model** is:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

But we estimate **only**:

$$Y_i = \alpha_0 + \alpha_1 X_{1i} + u_i \quad (\text{misspecified model})$$

What is $E[\hat{\alpha}_1]$? One can show:

$$E[\hat{\alpha}_1] = \beta_1 + \beta_2 \cdot \frac{S_{X_1 X_2}}{S_{X_1 X_1}}$$

OVB: When Is It a Problem?

The bias is non-zero only when **both** conditions hold:

Condition 1: $\beta_2 \neq 0$

The omitted variable X_2 actually affects Y .

(If it doesn't matter, leaving it out is harmless.)

Condition 2: $S_{X_1X_2} \neq 0$

The omitted variable X_2 is correlated with the included X_1 .

(If they are uncorrelated, OLS for β_1 is still unbiased.)

Direction of bias: $\text{sign}(\text{Bias}) = \text{sign}(\beta_2) \times \text{sign}(r_{X_1X_2})$

β_2	$\text{Corr}(X_1, X_2)$	Bias direction
+	+	Upward (overestimate β_1)
+	−	Downward

OVV: A Motivating Example

Hypothetical scenario: We want to estimate the effect of advertising spend X_1 on sales Y .

The true model includes firm size X_2 (larger firms spend more **and** sell more).

```
1 set.seed(7)
2 n      <- 100
3 size   <- rnorm(n, 50, 10)      # firm size (omitted variable)
4 adspend <- 5 + 0.4*size + rnorm(n, 0, 5) # corr. with size
5 sales   <- 20 + 2*adspend + 3*size + rnorm(n, 0, 10) # true:  $\beta_1=2$ ,  $\beta_2=3$ 
6
7 # Short regression (omitting size)
8 coef(lm(sales ~ adspend))["adspend"]
```

```
adspend
5.099163
```

```
1 # Long regression (including size)
2 coef(lm(sales ~ adspend + size))["adspend"]
```

```
adspend
2.313567
```

Multiple Linear Regression

The MLR Model

The **Multiple Linear Regression (MLR)** model with k regressors:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \varepsilon_i$$

Symbol	Meaning
β_0	Intercept — mean of Y when all $X_j = 0$
β_j	Partial slope — change in mean Y per unit increase in X_j , holding all other regressors constant (<i>ceteris paribus</i>)
ε_i	Error term — same assumptions as SLR

Note

Statistics II — Linear Regression: Goodness of Fit & MLR

The key word in MLR is ***ceteris paribus*** (“all else equal”). β_j measures the effect of X_j **after controlling for**

OLS in MLR

OLS still minimises $SSR = \sum e_i^2$, but now in $k + 1$ dimensions.

The normal equations become a system of $k + 1$ linear equations — solved efficiently in matrix form:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

Note

You do not need to solve this by hand. In R, `lm(Y ~ X1 + X2 + ... + Xk)` handles everything. The interpretation principles are the same as SLR — only the *ceteris paribus* qualifier changes.

The Gauss–Markov theorem extends to MLR: under A1–A5 (appropriately generalised), OLS is still **BLUE**.

MLR: Interpreting Coefficients

Hypothetical example: Explain operating profit (Y) using both sales revenue (X_1) and number of employees (X_2).

```

1 set.seed(5)
2 n          <- 30
3 revenue2   <- round(runif(n, 100, 500), 0)
4 employees  <- round(runif(n, 10, 80), 0)
5 profit2    <- round(5 + 0.15*revenue2 + 0.8*employees + rnorm(n, 0, 12), 1)
6
7 mlr_model  <- lm(profit2 ~ revenue2 + employees)
8 coef(mlr_model)

```

(Intercept)	revenue2	employees
21.4612459	0.1290236	0.6126250

Interpretation:

- $\hat{\beta}_1 = 0.129$: one additional m.u. of revenue is associated with 0.129 m.u. more profit, **holding number of employees constant.**

MLR: Full Summary

```
1 summary(mlr_model)
```

Call:

```
lm(formula = profit2 ~ revenue2 + employees)
```

Residuals:

Min	1Q	Median	3Q	Max
-21.8947	-9.0708	-0.1219	6.0029	25.3186

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	21.46125	9.51476	2.256	0.0324	*
revenue2	0.12902	0.02048	6.299	9.66e-07	***
employees	0.61262	0.11789	5.196	1.80e-05	***

Adjusted R^2

Why Regular R^2 Is Not Enough in MLR

In MLR, adding any variable — **even a completely irrelevant one** — will weakly increase R^2 :

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} \Rightarrow R^2 \text{ can only increase as we add regressors}$$

This makes R^2 unreliable for comparing models with different numbers of regressors.

Note

Quick demonstration:

```
1 set.seed(9)
2 noise <- rnorm(30)           # completely irrelevant variable
3 m1 <- lm(profit2 ~ revenue2 + employees)
4 m2 <- lm(profit2 ~ revenue2 + employees + noise)
5 c(R2_without_noise = summary(m1)$r.squared,
6   R2_with_noise     = summary(m2)$r.squared)
```

Adjusted R^2

The **adjusted** R^2 penalises for the number of regressors k :

$$\bar{R}^2 = 1 - \frac{\text{SSR}/(n - k - 1)}{\text{SST}/(n - 1)} = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}$$

Properties:

- $\bar{R}^2 \leq R^2$ always
- $\bar{R}^2$ **can decrease** when an irrelevant variable is added
- $\bar{R}^2$ can be negative (very poor fit)

Use $\bar{R}^2$ when:

- Comparing models with **different numbers** of regressors
- Deciding whether adding a new variable improves the model

Adjusted R^2 : Demo

```

1 set.seed(9)
2 noise <- rnorm(30)
3 m1 <- lm(profit2 ~ revenue2 + employees)
4 m2 <- lm(profit2 ~ revenue2 + employees + noise)
5
6 c(Adj_R2_without = summary(m1)$adj.r.squared,
7   Adj_R2_with    = summary(m2)$adj.r.squared)

```

Adj_R2_without	Adj_R2_with
0.6286853	0.6145915

Adding the irrelevant `noise` variable **increases** plain R^2 but **decreases** $\bar{R}^2$ — correctly signalling that the additional variable does not improve the model.

Lecture Summary

Key Takeaways — Lecture 9

- **SST = SSE + SSR**: total variation = explained + unexplained.
- $R^2 = \text{SSE}/\text{SST}$: proportion of variation explained by the model. In SLR, $R^2 = r^2$.
- The **Gauss–Markov theorem** guarantees OLS is BLUE under A1–A5.
- **OVB** arises when a relevant variable is omitted **and** it is correlated with included regressors — OLS becomes biased and inconsistent.
- **MLR** includes multiple regressors. Each $\hat{\beta}_j$ is a **partial effect** — the effect of X_j holding all other variables constant.
- **Adjusted R^2** penalises for extra regressors — use it to compare models of different sizes.

Looking Ahead

Lecture 10 — Part III: Statistical Inference & Model Validation

- Are the estimated coefficients statistically different from zero? — ***t*-tests**
- Is the model as a whole significant? — the ***F*-test** (ANOVA table)
- Confidence intervals for β_j
- Checking the classical assumptions — residual diagnostics
- Normality, homoscedasticity, autocorrelation, and multicollinearity

Note

Reference: Newbold Ch. 11.6–11.8 (inference), Ch. 12.4–12.6 (MLR inference and diagnostics).